

Data Science 5030

Understanding Uncertainty

University of Virginia, Fall 2026
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Course description: “Provides an in-depth exploration of probabilistic and statistical methods used to understand, quantify, and manage uncertainty. Learn foundational concepts in probability and statistics, simulation techniques, and modern approaches to parameter estimation, decision theory, and hypothesis testing. Topics include parametric and nonparametric methods, Bayesian and frequentist paradigms, and applications of uncertainty in real-world problems.”

In this course, we explore concepts from probability and statistics with a modern set of computational tools. Our goal is to start with foundational concepts in mathematics, introduce modern practices that use simulation and resampling techniques as the foundation of inference, and fit a variety of models and quantify uncertainty in predictions. Very roughly, the question for the first half of the course is, “What can we learn from data?”, and the question for the second half of the course is, “How uncertain are our predictions?”

Logistics:

GitHub Repo: https://github.com/ds4e/understanding_uncertainty

Background Videos: <https://www.youtube.com/@datasci4e>

Section 001: Tu/Th, 9:30 AM to 10:45 AM, SDS 306

Final exam: Dec 11, 2:00 PM to 5:00 PM

Section 002: Tu/Th, 11:00 AM to 12:15 PM, SDS 406

Final exam: Dec 10, 9:00 AM to 12:00 PM

On Tuesdays, we will focus on reviewing and introducing concepts and working through examples and problems. The class will start with a short quiz on the material for the previous week.

On Thursdays, we will apply the concepts from the week to a lab. You are encouraged to collaborate in groups and share code. All submissions, however, are considered your work, and you are answerable for everything that you submit.

Deliverables and Grading: We will generally use Canvas to distribute assignments, submit work, and return grades. The main deliverables are:

1. Weekly labs on Thursday, due on subsequent Wednesday nights, worth 30%
2. Weekly problem sets due on subsequent Wednesday nights, worth 20%
3. Weekly quizzes on Tuesday covering the previous week, worth 10%
4. A midterm and final exam, each worth 20%

Screen Free Tuesday: There is strong evidence that unrestricted classroom laptop use reduces learning, including for nearby students (Fried 2008; Junco 2012; Sana, Weston & Cepeda 2013; Carter, Greenberg & Walker 2017; Glass & Kang 2019). On Tuesdays, when our

primary work is discussion and developing concepts, we'll therefore generally put our screens away. Thursdays are more computationally oriented.

Office Hours: Wednesday, 1:30 PM to 2:45 PM, Zoom or fourth floor of SDS. There is no TA: I will grade your work and answer questions.

Services and Software, Course Materials: We will use Git/GitHub for collaborating and version control and Python for computing. For a package manager, we recommend Miniconda or uv. For an IDE, we recommend VS Code. If you are having trouble, you can always fall back on Google Colab or use it as you set up your own computing solution.

Textbook: There is no required textbook. Slides/notebooks will be provided for every class, including data and examples of code. There are many textbooks on probability and statistics, including Wasserman (All of Statistics); Casella and Berger (Statistical Inference); Hogg, McKean and Craig (Introduction to Mathematical Statistics); and Hansen (Probability and Statistics for Economists). For a more computational version of these topics, there is Efron and Hastie (Computer Age Statistical Inference). A thorough reference on probability is Grimmett and Stirzaker (Probability and Random Processes) or Durrett (Probability: Theory and Examples).

Prerequisites: As prerequisites for admission to the MSDS program, students have completed courses in statistics, single-variable calculus, linear algebra, and programming. We provide review materials in the form of videos for students whose recall of these topics has depreciated.

Who is this class for? Everyone is welcome to participate in this class, of any age, culture, gender, language or geographic heritage, learning and physical abilities, political or social beliefs, race or ethnicity, religious or spiritual beliefs, sex, and social or economic class. Conversely, everyone participating in this class is expected to respect the dignity and humanity of their peers. Please feel free to tell me how you would like to be addressed.

Academic Integrity: (Wording suggested by the administration) "I trust every student in this course to fully comply with all of the provisions of the University's Honor Code. By enrolling in this course, you have agreed to abide by and uphold the Honor System of the University of Virginia. All graded assignments must be pledged, including homework and exams. All suspected violations will be forwarded to the Honor Committee. Please let me know if you have any questions regarding the course Honor policy. If you believe you may have committed an Honor Offense, you may wish to file a Conscientious Retraction by calling the Honor Offices at (434) 924-7602."

Generative Artificial Intelligence: You can use generative AI (e.g. Claude, Codex, Copilot, etc.), but if you use these tools to generate significant chunks of code (not just advanced auto-completion or to get ideas), acknowledge it in the commented code or written answer that you submit and be prepared to explain the output. You are answerable for the work you submit.

Absences, late work, extra credit: I don't take attendance and you don't have to explain your absence from class. If your work will be submitted late, please contact me in advance to explain. There is no extra credit.

Students with Accommodations: If you are entitled to any accommodations, please get in touch with me as soon as possible.

Learning Objectives:

- Model random phenomena with the foundational probability space model
- Describe observed randomness with empirical cumulative distribution functions and kernel density plots
- Represent uncertainty mathematically with cumulative distribution functions and probability density functions
- Compute conditional and unconditional expectations, and explain how they relate to conditional expectation functions and other predictors
- Simulate random phenomena computationally
- Analyze sampling distributions empirically using the bootstrap
- Compare outcomes between multiple populations
- Understand multidimensional densities and distributions and the independence property
- Fit and simulate random processes using Markov Chains
- Use the likelihood to construct a relationship between data and parameters
- Use models to make probabilistic predictions
- Apply Bayes' rule to elementary inference problems to learn about parameters from data

Grading Scale:

- 97-100 A+, 93-96.9 A, 90-92.9 A-
- 87-89.9 B+, 83-86.9 B, 80-82.9 B-
- 77-79.9 C+, 73-76.9 C, 70-72.9 C-
- 67-69.9 D+, 63-66.9 D, 60-62.9 D-
- <60 F

Class Schedule: This is an outline of the class and subject to change:

Concepts Day		Applications Day	
08/25/26	Introduction	08/27/26	Visualization
09/01/26	Vectors and Inner Product	09/03/26	Cosine Similarity, TF-IDF
09/08/26	Random Variables, Expectation	09/10/26	Statistical Decision Theory
09/15/26	Categorical Variables and ECDF	09/17/26	CDF and PDF
09/22/26	KDE	09/24/26	Hazard and Survival
Sampling Distribution and			
09/29/26	Bootstrapping	10/01/26	CLT and Inference
10/06/26	<i>Fall Break</i>	10/08/26	Exam
10/13/26	Joint Densities, Independence	10/15/26	Conditional Density/Expectation
10/20/26	Empirical Markov Chains	10/22/26	Simulating Markov Chains
10/27/26	The Likelihood Function	10/29/26	Maximum Likelihood Estimation
11/03/26	<i>Election Day</i>	11/05/26	Linear Regression
11/10/26	Logistic Regression	11/12/26	Gradient Descent
11/17/26	The Bayesian Approach	11/19/26	Conjugate Priors
11/24/26	Sampling the Posterior	11/26/26	<i>Thanksgiving</i>
12/01/26	Advanced topics	12/03/26	Advanced Topics
12/08/26	Review		